

The Problem: Mixed Traffic in Vietnam

- AV motion planners assume Western traffic: marked lanes, rule-abiding drivers.
- In Vietnam, >90% of vehicles are motorcycles that actively:
 - lane-split and squeeze through sub-meter gaps,
 - cut in at short range and ignore signals.
- Standard planners react too late → unsafe in these scenes.
- Goal: a planner that ANTICIPATES motorcycle cut-ins, on the real VinUni / Ocean Park road.

Method: APEX predictive risk-aware planner

fuses car-following + reactive avoidance + sampling; fixes each flaw

- 1) Predict every motorcycle over a 4 s horizon + reachability envelope
 - anticipates a cut-in before it starts (vs blind constant-position).
- 2) Hard footprint-clearance margin to all motos over the whole horizon
 - collision-free by construction (fixes the point-box flaw).
- 3) Junction-yield speed-cap for roadside / crossing motos
 - solves crossings (0/4) where every other planner fails.
- 4) Multi-horizon trajectories → brake hard when needed, full speed when clear (fixes IDM/conservative over-braking).
- 5) Speed-maximizing objective under the safety constraint
 - most efficient among collision-free planners.
- Runs on the real VinUni CommonRoad map + recorded GPS.

Results: 52 scenarios x 6 planners

only conservative & APEX are crash-free; APEX is the efficient one

Planner	Collisions	<0.3 m fails	Min-clr	R_T
IDM (Treiber'00)	5/52	7/52	5.73	1.57
ORCA/VO (vdB'11)	1/52	1/52	2.02	1.34
baseline Frenet	28/52	30/52	1.24	1.27
conservative	0/52	0/52	13.96	1.86
moto-aware (prev ours)	13/52	14/52	2.31	1.33
APEX (predictive, OURS)	0/52	0/52	2.16	1.43

APEX vs the rest:

- 0 collisions across all 52 → -100% vs baseline (28) and vs our previous (13)
- 23% faster than the only other crash-free planner (R_T 1.43 vs 1.86)
- beats published IDM & ORCA baselines; crossings solved 0/4